Bitcoin, Cryptocurrencies, and Blockchain Technology

1. Bitcoin & Cryptocurrencies
2. Blockchain Technology
3. Security and Privacy in Bitcoin and Cryptocurrencies

CRYPTOGRAPHY AND SECURE DEVELOPMENT
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Outline

1. Bitcoin & Cryptocurrencies
   - Decentralised Payment System: A Big Picture
   - Cryptography in Bitcoin
   - Bitcoin Protocol: an Overview
2. Blockchain Technology
   - Bitcoin Addresses
   - Bitcoin Distributed Ledger: a Blockchain
   - Transactions in Bitcoin and Blockchain
   - Consensus Mechanism: Proof-of-Work
3. Security and Privacy in Bitcoin and Blockchain

Security and Privacy in Bitcoin

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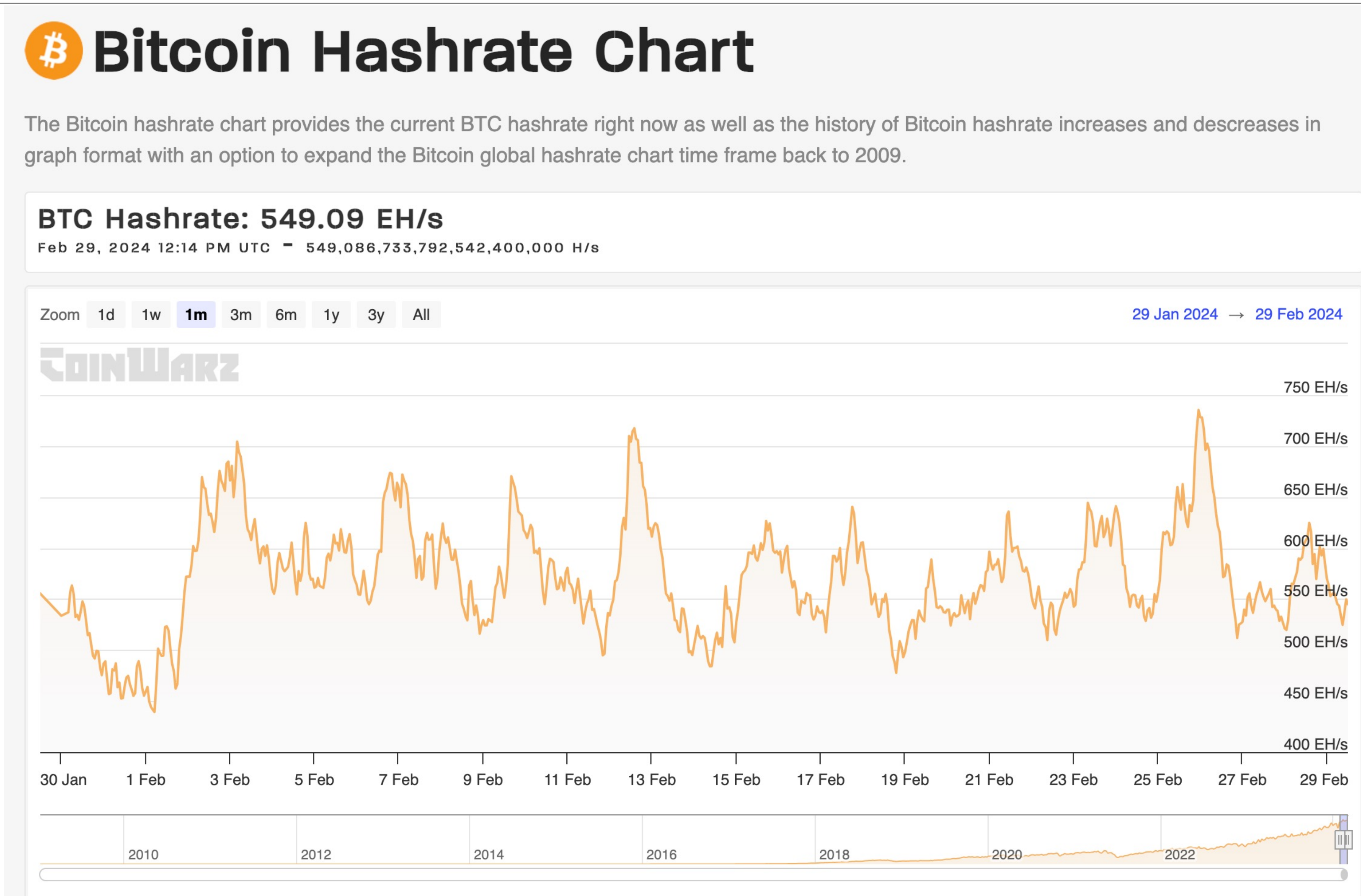
Security in Bitcoin

- Authentication → Public Key Crypto & Digital Signatures
  - Am I paying the right person? Not some other impersonator?
- Availability → Broadcast messages to the P2P network
  - Can I make a transaction anytime I want?
- Integrity → Digital Signatures & Cryptographic Hash
  - Is the coin double-spent?
  - Can an attacker reverse or change transactions?
- Confidentiality → Pseudonymity
  - Are my transactions private? Anonymous?

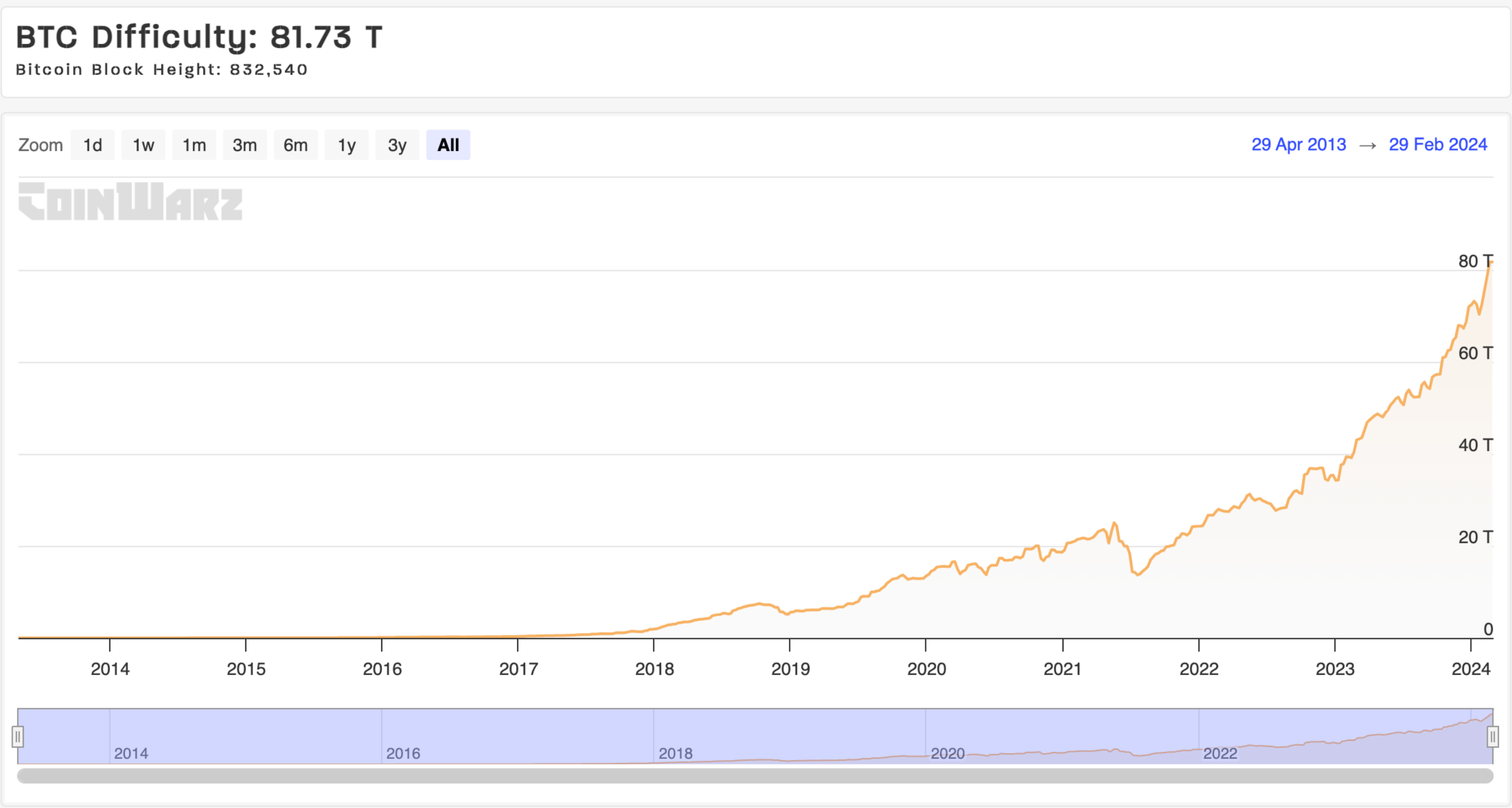
Mining: The Guardian of the Bitcoin Network

- Mining, beyond generating new bitcoin, validates and embeds transactions into the blockchain.
- This Proof-of-Work process is pivotal, requiring substantial computational resources and underpinning the network's security.

Proof-of-Work: Bitcoin Hash-rate



Proof-of-Work: Bitcoin Difficulty



Transaction Finality

- Finality is the guarantee that past transactions can never change.
- Bitcoin system only offers probabilistic transaction finality - that transactions are not immediately final but become so eventually.

Blockchain	Consensus Algorithm	Average time to mine a block	Average Time to Finality
Bitcoin	PoW	10mins	60mins (6 confirmations)
Ethereum	PoW	15 secs	6 mins (25 confirmations)
Nxt	DPoS	1 min	720 mins (720 blocks)

Double-Spend in Bitcoin

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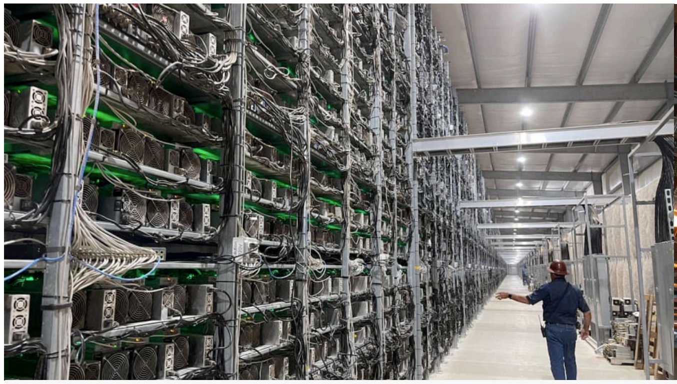
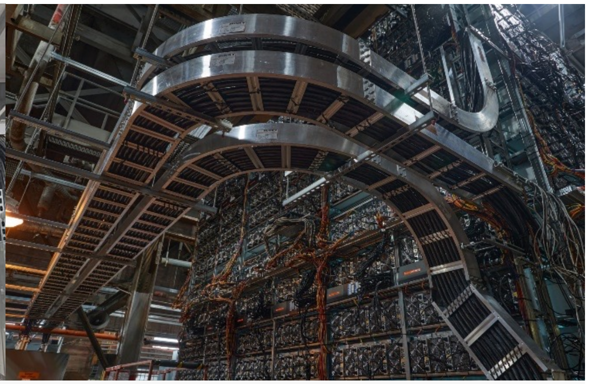
Why Proof of Work?

- Why do we need to use an extremely resource-intensive computation for "Nonce"
  - F.Y.I: Bitcoin currently consumes around 110 Terawatt Hours per year — 0.55% of global electricity production* (2021)
  - From CPU to GPU to ASICs
- Integrity: To prevent from transaction alteration/reverse and Double-spend.
  - Intuitively, to change/reverse Tx, a malicious miner needs to re-compute nonce for several blocks while racing with other honest miners for new block → nearly impossible

* https://ccaf.io/cbeci/index

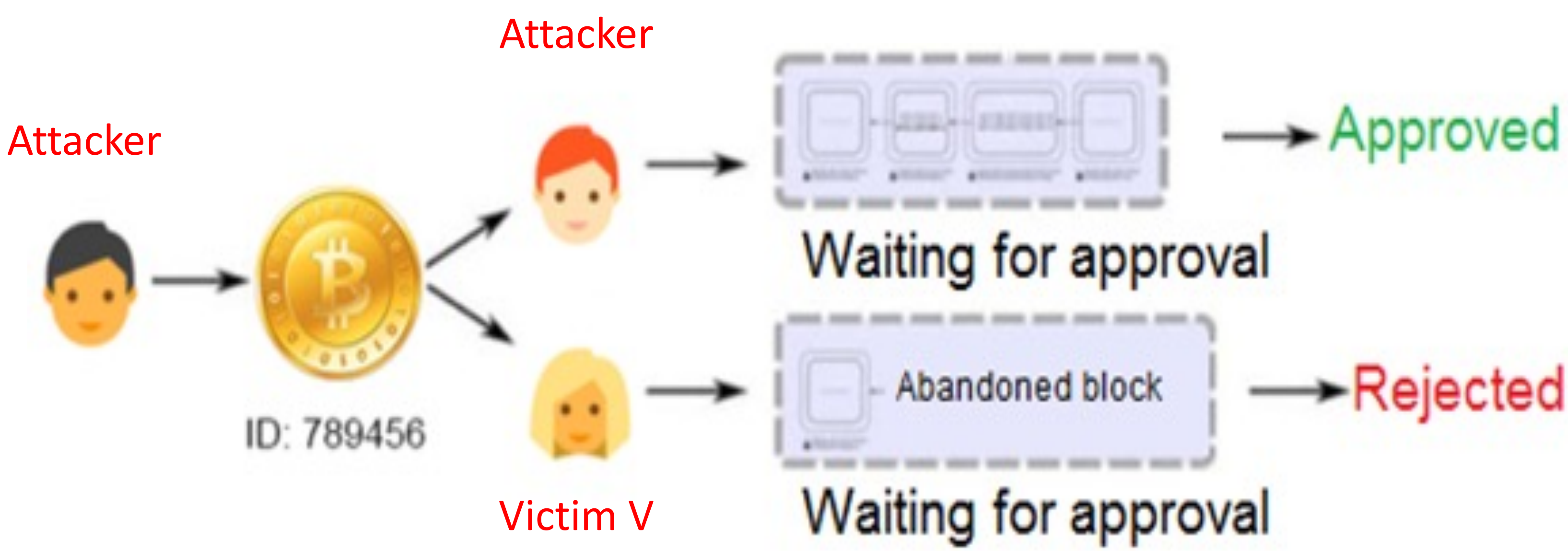
Why Proof of Work?





Double-Spend in Bitcoin

- Double-spending problem is the successful use of the same funds twice.
- Here is the strategy for attacker performing double-spend attack:



Double-Spend in Bitcoin

- Attacker makes a transaction transferring its money to victim V (transaction X).
- Attacker makes another transaction transferring this money to its (another) address (transaction Y)
  - Secretly mining using the block that includes transaction Y.
- Wait for the transaction X to be confirmed and included in a new block → Victim V hands over his goods, sure that the money is finally appropriated to him.

Double-Spend in Bitcoin

- The attacker then broadcasts the block he mined which contains the transaction Y → creates a fork
- And continues to mine this alternative branch (i.e., fork) and successfully mine another block appended to this fork → broadcast to the network
- Since the new branch is longer than all other known, it will be considered valid, and BTC transfer to the victim V will be replaced by sending coins to the attacker.

Double-Spend in Bitcoin

- Double-spend attack: a race between attackers (malicious miners) and honest miners.
- Basically, to be successful in the double-spend attack, the attackers need to control > 50% computation power of the whole Bitcoin network (i.e., 51% attack)
  - This is practically impossible.

Decentralisation in Bitcoin

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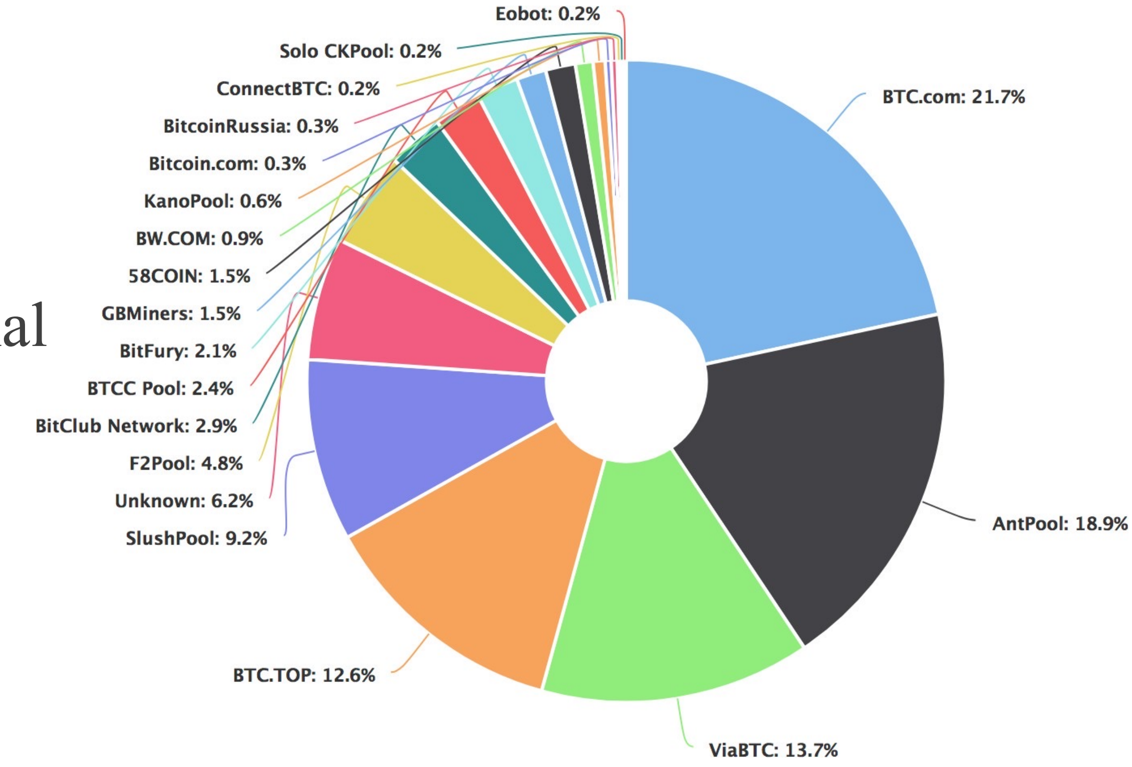
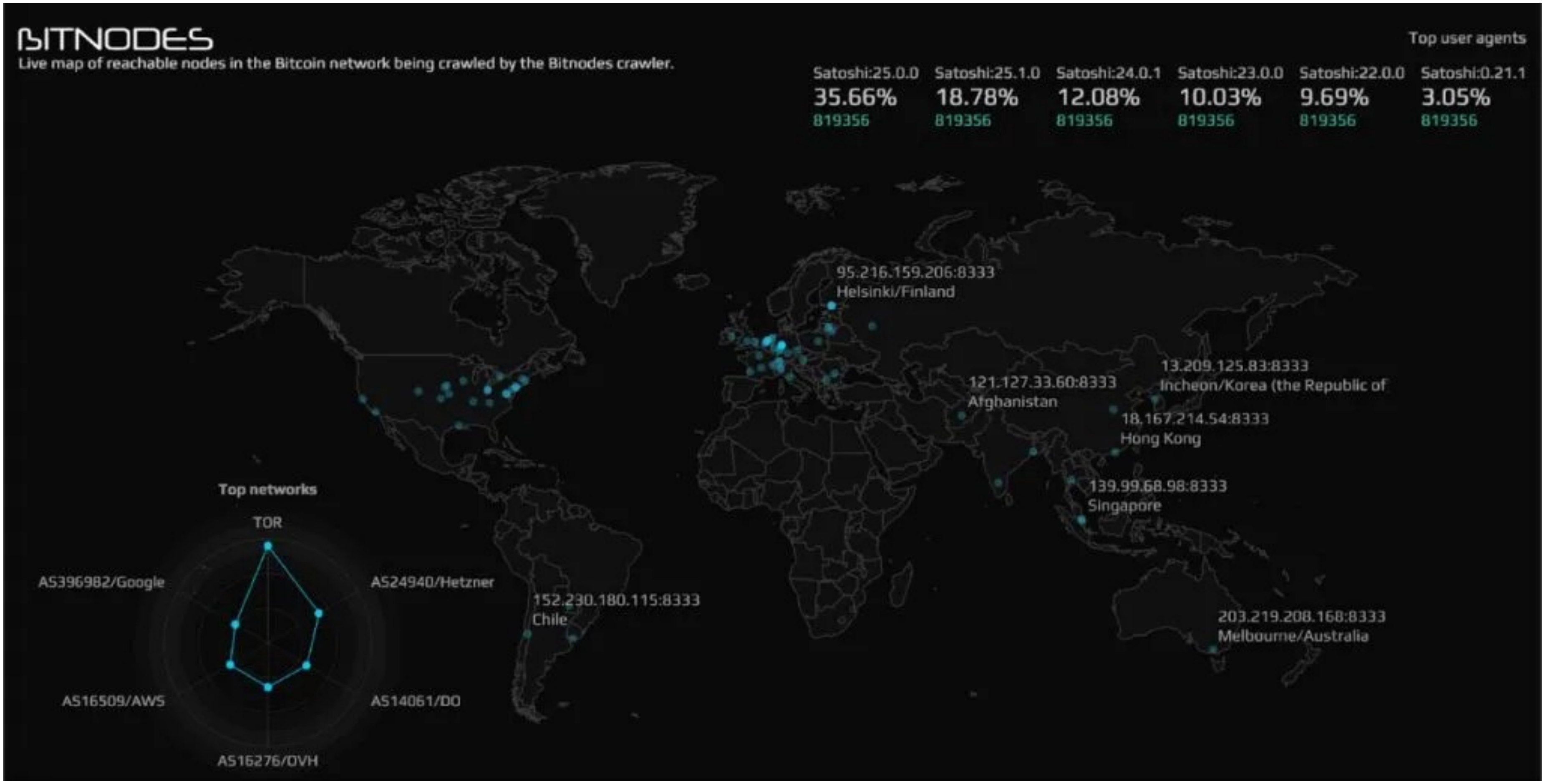
Bitcoin Mining Pools

- Mining Pools:
  - Probability of finding a block alone is very small
  - Unite in Mining pools
  - Payout is done proportional to the work
  - What if the mining pools collude and carry out double-spend attack?



Full nodes across the Globe



Tens of thousands of nodes spread across the globe — Source: Bitnodes.io

Privacy in Bitcoin: Pseudonymity

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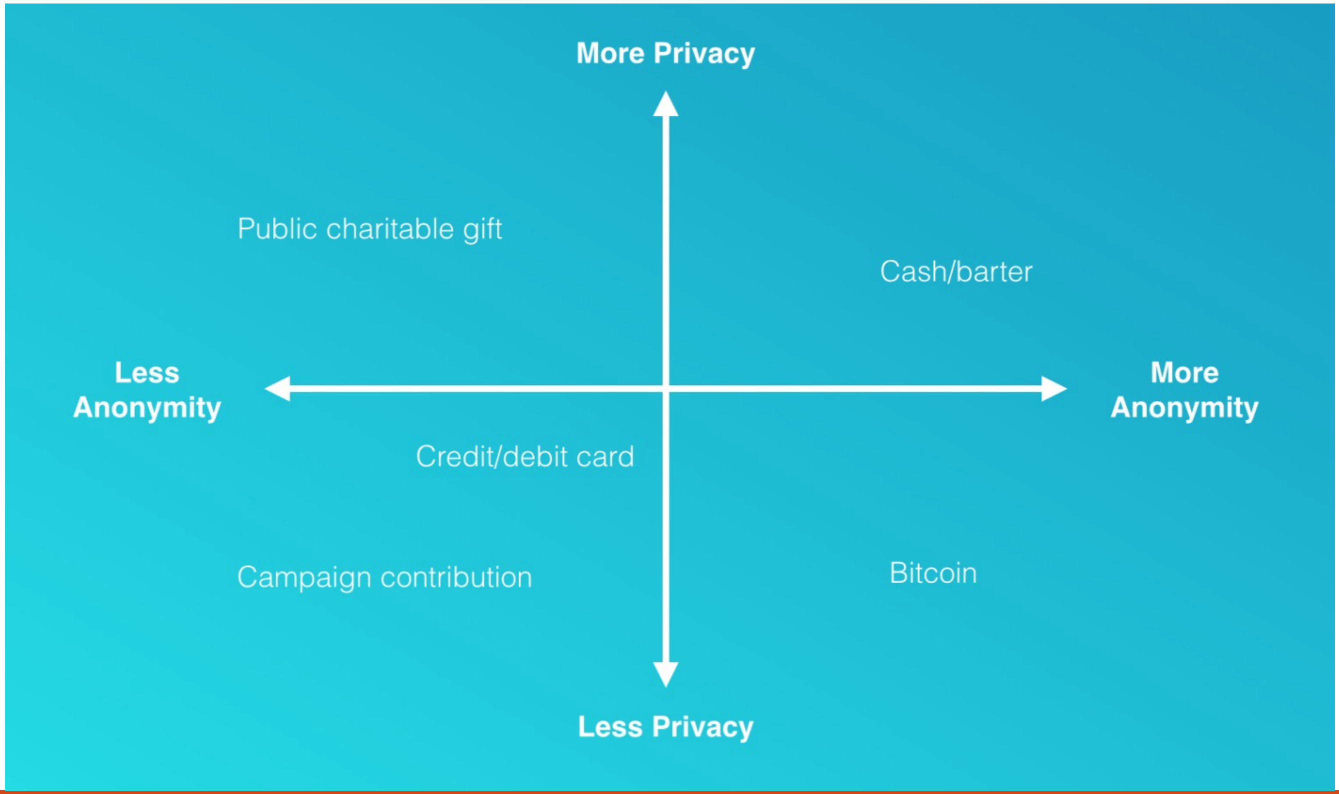
Definitions

- A transaction is "anonymous" if no one knows who you are.
- A transaction is "private" if what you purchased, and for what amount, are unknown



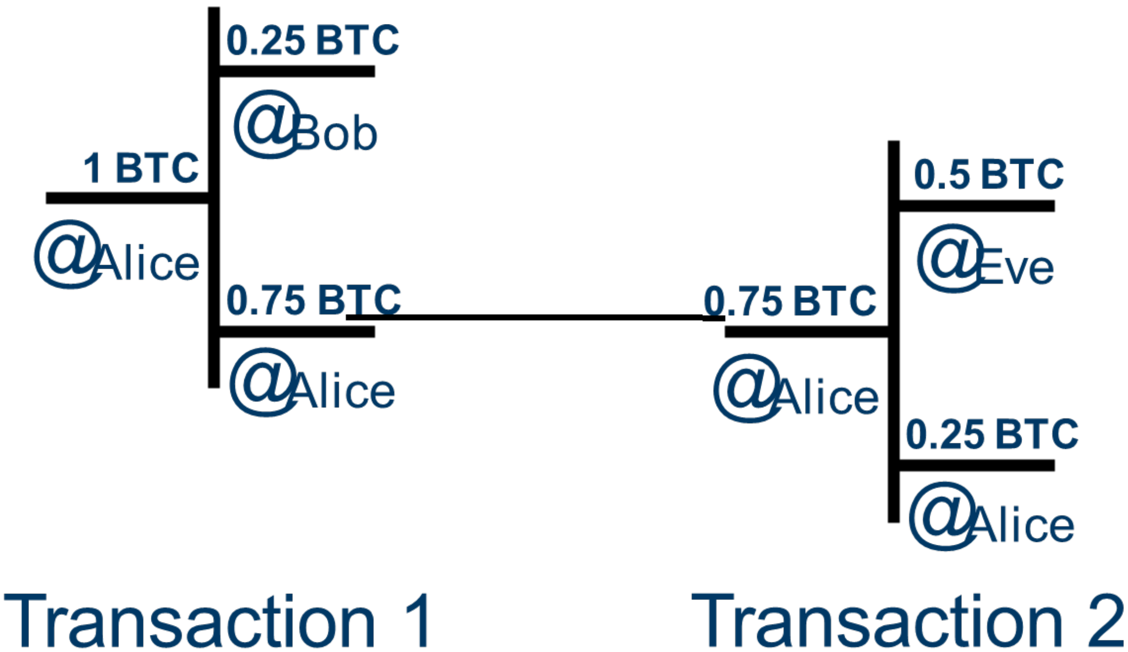
Bitcoin Privacy: Pseudonymity

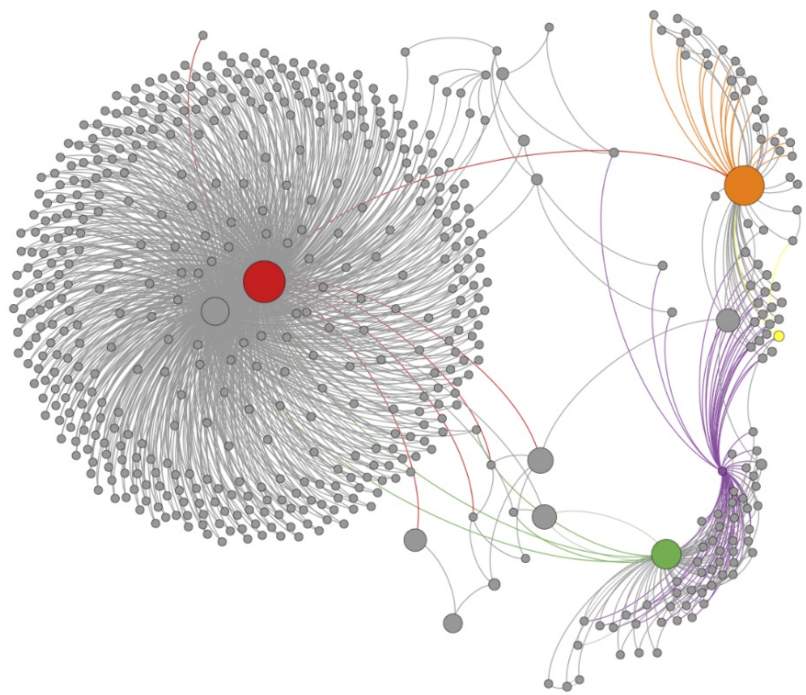
- Bitcoin is assumed to be anonymous
  - As Bitcoin addresses are as random string of characters (hash of hash of a public-key)
- But not private
  - Identities are not revealed in the blockchain – but every transaction is visible in the blockchain.
- Anonymity is attractive – but also a challenge for financial regulation that seeks to prevent money laundering

Bitcoin Privacy: Pseudonymity

- Even Bitcoin is anonymous, there is a way to de-anonymise
  - This is called "transaction graph analysis"
- What's wrong here?
  - Alice BTC relates to each other due to the definition of transactions (e.g., multiple inputs, change addresses)
  - Combined with other side-information or low-layer network information (e.g., IP addresses) in the Bitcoin network
  - There is a change of figuring out the real identity of BTC owners.

Bitcoin Privacy: Pseudonymity





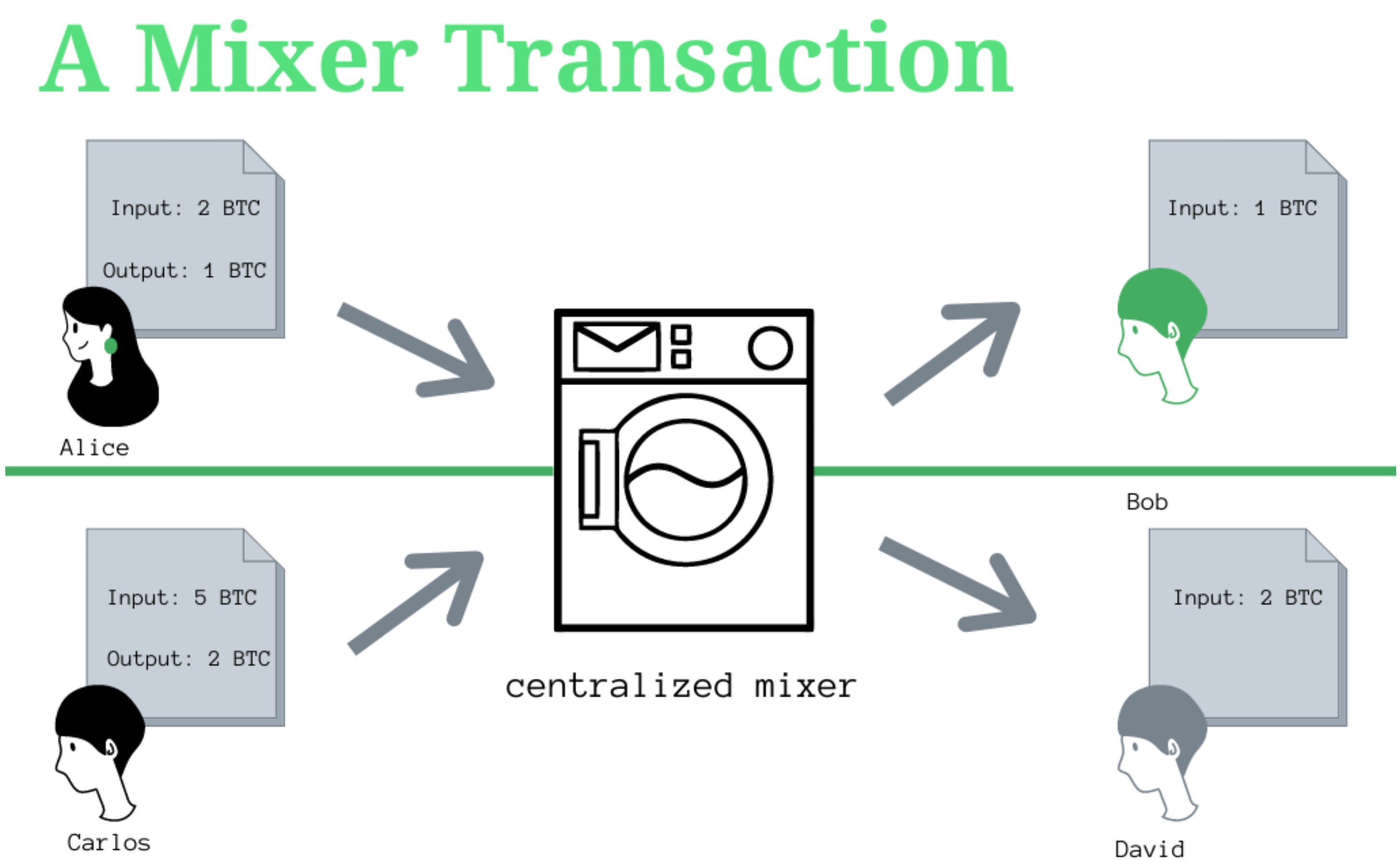
Bitcoin Fungibility

- What Is Fungibility?
  - Fungibility is a property of goods whose units are interchangeable.
  - A good is fungible if one unit of the good always carries the exact same value as all other units of the good.
  - Fiat currency: GBP notes?
  - Is Bitcoin Fungible?
- Fungibility is critical to preserving Bitcoin's censorship resistance and privacy.
  - Goods which are not fungible or divisible serve as poor monetary goods. Bitcoin has infinite divisibility and strong fungibility.

Bitcoin Privacy: Solutions

- Some solutions:
  - Mixer, JoinCoin, k-anonymity Privacy, N-anonymity Privacy, Zero-Knowledge-Proof (ZKP)
  - This is the task for you to explore!

Bitcoin Privacy: Solutions
Bitcoin Mixing Service



Alice wants to send 1 BTC to Bob. Carlos wants to sent 2 BTC to David. Neither wants the transactions tied to their addresses. They send the BTC to a centralized mixing service which shuffles a pool of BTC and sends different coins to Bob and David.

Bitcoin Privacy: Solutions
Coin-Join